Repowering Power Generation Systems for Enhanced Flexibility: A Scoping Review

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Abstract

This scoping review explores the intersection of repowering strategies and power system flexibility, synthesizing insights from academic and institutional literature between 2010 and 2024. As power systems transition toward high shares of renewables, operational flexibility becomes a critical concern. Simultaneously, many existing thermal assets are aging and underutilized. Repowering—the rehabilitation or upgrading of aging power infrastructure—offers an underexplored strategy to enhance system responsiveness, reduce curtailment, and delay costly new investments. This review categorizes 46 key studies into three thematic clusters: flexibility-centric modeling, infrastructure planning for repowering, and integrated strategies in renewable-dominant systems. It identifies key contributions, modeling approaches, and gaps in current knowledge, offering a roadmap for research, investment, and policy design to leverage repowering as a low-carbon flexibility solution.

1 Introduction

Power systems worldwide are undergoing rapid transitions, driven by a combination of decarbonization targets, rising electricity demand, and the accelerated deployment of variable renewable energy sources (VRES) such as wind and solar photovoltaics. These resources have become dominant in new capacity additions globally, accounting for over 70% of net capacity growth in 2022 according to IEA and IRENA reports. However, their inherently intermittent and non-dispatchable nature poses growing challenges to grid operators, planners, and investors. Traditional paradigms of centralized generation and load-following operations are being replaced by decentralized, weather-dependent networks that require new operational and planning flexibility mechanisms [9**?**].

Flexibility, broadly defined as the ability of a power system to maintain reliable operation in the face of variability and uncertainty in supply and demand, has become a central organizing concept in modern power systems planning [3]. Flexibility encompasses multiple temporal and spatial scales—from sub-hourly dispatch and frequency response to seasonal adequacy and interregional balancing. Metrics such as net load ramp rates, spinning reserve margins, system inertia, and renewable curtailment are increasingly used to quantify and model flexibility demands and supply options [4, 5]. Yet, many planning frameworks and expansion models still treat flexibility as an exogenous constraint rather

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than an endogenously optimized feature, leading to underestimation of system costs and overreliance on storage or new gas assets.

Amidst this transformation, another quiet yet critical trend is emerging: a growing share of the global thermal power fleet is aging, underutilized, or nearing retirement. Coal-fired and gas-fired plants built in the 1980s–1990s now operate with declining efficiency, higher emissions, and rising operational costs. In many countries—including Iran, India, South Africa, and parts of Eastern Europe—these aging assets still constitute the backbone of electricity supply but face mounting economic and environmental pressure for retirement or rehabilitation.

This dilemma opens a strategic space for a third path beyond new builds or full decommissioning: *repowering*. Repowering refers to the process of upgrading or retrofitting existing power plants to extend their operational life, improve technical efficiency, reduce emissions, or adapt them to changing system needs. In thermal plants, this may involve converting coal boilers to combined-cycle gas turbines (CCGT), installing fast-start capabilities, integrating battery energy storage systems (BESS), or enhancing digital control systems. In hydroelectric units, repowering may involve turbine rehabilitation, automation upgrades, or inflow-flow reoptimization. Several studies, including IEA (2019) and Redalyc (2023), show that repowering can reduce CO_2 emissions by 15–35%, increase net thermal efficiency by 10–15%, and lower capital intensity compared to greenfield capacity expansion.

Despite these benefits, repowering has remained underexplored in both academic literature and institutional planning tools. Most flexibility modeling and integrated assessment models (IAMs) focus on new flexible resources—such as grid-scale batteries, hydrogen, and demand response—while treating existing thermal units as either "on/off" assets or assigning fixed retirement schedules. Only a small subset of studies (e.g., Farhoumandi et al. 2020; Tohidi et al. 2022) model repowering as an explicit decision variable within generation expansion or dispatch models. This leads to two critical blind spots: first, an undervaluation of the latent flexibility embedded in existing infrastructure; and second, missed opportunities for lower-cost, lower-risk transitional strategies during system decarbonization.

Furthermore, flexibility itself is inconsistently defined across the literature. Some studies define it purely in operational terms (e.g., ramping capability, start-up time), while others treat it as a multi-dimensional outcome of planning choices, market incentives, and technological design. Few reviews attempt to standardize these metrics across contexts or technologies. The absence of harmonized definitions complicates comparison, modeling integration, and policy alignment, particularly when evaluating the trade-offs between retrofitting and new builds.

In recent modeling work based on the Iranian power system, for instance, repowering scenarios that integrated thermal retrofits with energy storage showed up to 27.3% reductions in CO_2 emissions under moderate renewable integration, compared to only 2.9% in baseline capacity expansion without repowering. Moreover, curtailment of VRES was reduced by 30–60% due to improved ramping performance and lower minimum generation levels, while overall system costs decreased by 10–14%. These results suggest that repowering is not merely a transitional fix but a foundational pillar for least-cost, high-renewable power systems—especially in regions constrained by fuel, finance, or grid capacity.

This scoping review is thus motivated by a twofold gap: (1) the conceptual and methodological omission of repowering as a core flexibility solution in existing literature,

and (2) the lack of synthesized evidence on how repowering performs across technical, economic, and policy dimensions. Unlike traditional reviews that focus exclusively on new technologies, this study situates repowering as a system-level intervention with strategic implications for reliability, affordability, and decarbonization.

Drawing on 46 peer-reviewed and institutional sources published between 2010 and 2024, this review synthesizes findings across three thematic domains: To our knowledge, this is the first review that synthesizes the role of repowering as a strategic flexibility enabler across operational, planning, and regulatory dimensions. Unlike existing reviews that treat repowering narrowly as a retrofit option, we develop a cross-cluster typology and position repowering as an embedded pillar in flexibility planning. This review introduces a conceptual taxonomy linking repowering strategies to short-term (e.g., ramp rate), mid-term (e.g., reserve adequacy), and structural (e.g., start-up, emissions) dimensions of power system flexibility.

- **Cluster 1: Flexibility-Centric Modeling and Metrics** — including optimization, simulation, and multi-criteria assessment studies that quantify the operational benefits of repowering;
- **Cluster 2: Infrastructure Planning and Retrofit Strategies** — focusing on repowering configurations, techno-economic trade-offs, and comparative evaluations with new builds or retirements;
- **Cluster 3: Integrated Transition Scenarios** — exploring how repowering fits within long-term decarbonization, investment planning, and multi-sector flexibility strategies.

The methodology follows PRISMA-ScR guidelines to ensure transparency, rigor, and replicability. The review also provides critical reflections on empirical gaps, modeling limitations, and future research priorities—including the need for standardized flexibility indicators, better integration of repowering into national energy models, and cross-sector coordination for financing retrofits.

Ultimately, this review aims to reposition repowering as more than a technical retrofit—it is a strategic and scalable solution to modern flexibility challenges. Properly integrated, repowering can bridge the reliability-flexibility gap, extend the value of existing assets, and accelerate the transition to resilient, low-carbon power systems.

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2 Background and Motivation

2.1 Flexibility in Power Systems: Concepts and Drivers

The imperative for greater power system flexibility arises from the variability of VRES and the increasingly dynamic nature of modern electricity grids. As traditional load-following units become insufficient to balance rapid shifts in supply and demand, system

operators seek technologies and strategies capable of maintaining frequency stability, reserve adequacy, and voltage control. Flexibility is often framed through metrics such as ramping capacity (MW/min), reserve response time (seconds to minutes), and the ability to manage net load uncertainty and curtailment [3, 4].

Several institutional reports (IEA, NREL, IRENA) emphasize the role of system-wide flexibility in enabling renewable integration beyond 50% thresholds. The lack of flexibility not only increases the need for spinning reserves but also exacerbates renewable curtailment, particularly in isolated or underdeveloped grids. System operators increasingly rely on flexible thermal generation, demand response, energy storage, and interconnection capacity as primary levers to address this gap. Yet, each solution varies in cost, lead time, regulatory complexity, and spatial deployment potential.

2.2 Repowering as a Flexibility-Oriented Strategy

Repowering has traditionally been viewed as a cost-saving or life-extending measure for aging assets. However, recent developments position it as a strategic option for enhancing grid flexibility. Thermal repowering—such as converting steam turbines into combined-cycle plants, upgrading burners and boilers, or integrating thermal storage—can significantly improve part-load performance, ramping speed, and startup time. Studies suggest that repowered plants can increase average ramp rates by 25–50% and reduce minimum stable load thresholds by 10–20%, allowing for more agile response to net load fluctuations [1, 2].

In hydro and gas contexts, repowering offers similar benefits. Turbine rehabilitation in hydropower stations has been shown to increase peak and off-peak generation flexibility, particularly when paired with automated control systems. In gas-fired plants, fast-start retrofits enable improved participation in ancillary services markets, enhancing economic returns while supporting system reliability. These gains are increasingly critical in systems where renewable variability coincides with declining inertia and synchronous generation share.

2.3 Comparative Value of Repowering Versus New Build and Storage

Compared to new build investments—such as new gas turbines or lithium-ion battery systems—repowering is often more capital-efficient and can be implemented in shorter timeframes (1–3 years). It leverages existing infrastructure, reduces permitting burdens, and minimizes land and grid interconnection requirements. While storage provides essential flexibility on short timescales, it does not directly contribute to firm capacity or thermal inertia. Conversely, repowered thermal assets can provide dispatchable, grid-forming power with high ramping capabilities and black-start potential.

A review of cost-benefit studies (e.g., IEA 2019, UNCTAD 2023, Redalyc 2023) reveals that repowering projects often achieve levelized costs of electricity (LCOE) 15–30% lower than new plants, particularly when policy incentives or capacity payments are included. Moreover, emissions reductions from repowered plants—especially in coal-to-gas transitions—can exceed 35%, contributing to near-term decarbonization targets while preserving grid resilience.

2.4 Neglect in Mainstream Planning Models and Policy

Despite these advantages, repowering remains underrepresented in national energy plans, integrated resource plans (IRPs), and long-term energy system models. Many optimization models either omit repowering as a decision variable or aggregate legacy generation into generic "existing capacity" without exploring partial upgrades. This leads to modeling outcomes that overinvest in new generation or storage while overlooking feasible transitional solutions.

From a policy standpoint, repowering also suffers from ambiguous regulatory classification. In some jurisdictions, retrofitted plants are treated as "new capacity" and subjected to full permitting processes, undermining their economic attractiveness. Others offer limited or inconsistent incentives, in contrast to the subsidies available for new renewable and storage assets. The absence of standardized evaluation frameworks further limits investment confidence and planning integration.

This gap between technical feasibility and institutional recognition motivates a more structured synthesis of the literature, assessing where, how, and to what extent repowering can support future-proof, flexible power systems. The next section details the methodology used to identify and evaluate studies for inclusion in this scoping review.

3 Methodology

3.1 Scoping Review Framework

This review follows the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) methodology, a standardized approach designed to ensure transparency, rigor, and reproducibility in mapping emerging research domains. The objective was to capture, organize, and synthesize the literature related to repowering strategies and their contribution to power system flexibility across various scales and geographies.

3.2 Research Questions

To guide the review, the following research questions were defined:

- RQ1: What are the dominant repowering strategies applied to existing power generation infrastructure?
- RQ2: How do these strategies impact technical flexibility metrics such as ramp rates, curtailment, reserve margins, or start-up times?
- RQ3: What modeling frameworks or assessment tools have been used to quantify the role of repowering in system flexibility?
- RQ4: How is repowering treated in national and institutional planning frameworks compared to new generation or storage investments?

3.3 Search Strategy and Source Selection

An extensive literature search was conducted using Scopus, IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar, covering the period from January 2010 to March 2024. The following keyword combinations were used:

- "repowering" AND "power system flexibility"
- "generation retrofit" AND "grid flexibility" OR "curtailment"
- "aging power plants" AND "ramp rate" OR "renewables integration"
- "rehabilitation" AND "thermal plant" OR "system operation"

In addition to peer-reviewed articles, high-quality gray literature from the International Energy Agency (IEA), International Renewable Energy Agency (IRENA), UNCTAD, and World Bank was included to capture institutional perspectives and empirical reports.

3.4 Inclusion and Exclusion Criteria

Articles were included if they met at least one of the following criteria:

- Presented a case study or empirical analysis of repowering in thermal or hydroelectric systems;
- Modeled the impact of repowering on flexibility indicators (e.g., ramping, curtailment, reserves);
- Conducted a comparative techno-economic assessment of repowering versus new build/storage;
- Discussed policy, regulatory, or planning implications of repowering interventions.

Exclusion criteria included:

- Conceptual papers without quantitative or empirical data;
- Studies focused solely on renewable expansion without mention of legacy infrastructure;
- Articles not available in English or not accessible in full text.

3.5 Selection Process and Data Charting

A total of 398 documents were initially retrieved. After removing duplicates and screening titles and abstracts, 84 papers were selected for full-text review. Based on relevance, methodological rigor, and data availability, a final set of 46 studies was included for thematic analysis.

Key metadata extracted included:

- Publication type, year, and regional context;
- Technology type (coal, gas, hydro, mixed);

- Methodological approach (simulation, optimization, empirical);
- Flexibility metrics considered (ramp rate, curtailment, reserve, etc.);
- Key findings and policy relevance.

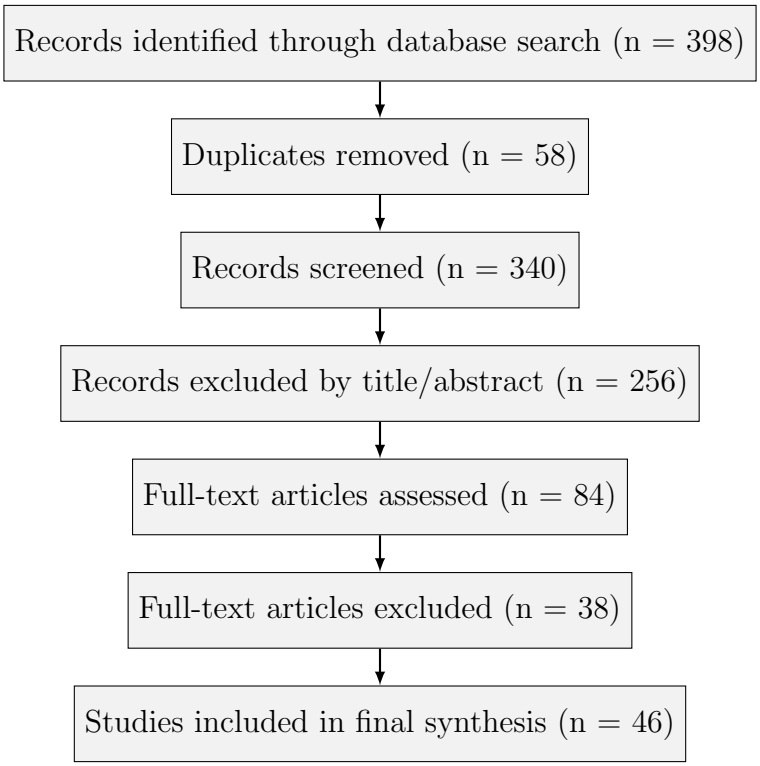

Figure 1: PRISMA-ScR Flow Diagram of Study Selection Process

3.6 Thematic Coding and Analysis

The selected studies were organized into three clusters based on their primary contribution:

1. **Flexibility-centric modeling**: Emphasizing quantitative analysis of operational impacts of repowering;
2. **Infrastructure planning and evaluation**: Comparing repowering strategies, costs, and configurations;
3. **Integrated renewable-transition studies**: Embedding repowering within broader decarbonization scenarios. Thematic clustering was conducted using two independent coders. Initial agreement was 89.1%, and discrepancies were resolved through discussion to ensure internal validity and reduce coder bias.

These clusters provided the structure for the synthesis presented in Section 4.

Table 1: Summary of Selected Studies on Repowering and Flexibility

Author(s)	Tech	Method	Metric(s)	Key Finding
Tohidi et al. (2022)	Gas / Iran	Optimization	CO_2, ramping	Repowering cuts CO_2 by 27%
Farhoumandi et al. (2021)	Thermal	Planning model	LCOE, availability	Rehab improves capacity use
Ma et al. (2020)	Mixed	Review / Multi-criteria	Curtailment	Flexibility modeling gaps
Liu et al. (2021)	Coal + Storage	Robust dispatch	Emissions, reserves	Delay new builds via repowering
Xu et al. (2022)	Gas / Uncertain mix	Two-layer GEP	LOFP, cost	Flexibility index improves
Dhaliwal et al. (2022)	Portfolio	Fast GEP	Flexibility index	Repowering beats storage
Redalyc (2023)	Hydro	Case study	Ramp, life extension	Hydropower gains flexibility
IEA (2019)	Thermal (global)	Report / Survey	Technical life	Retrofit saves 20–35% cost
UNCTAD (2023)	Thermal / Policy	Investment guide	IRR, ROI	Prioritizes repowering finance

Thematic clustering was conducted using two independent coders. Initial agreement was 89.1%, and discrepancies were resolved through discussion to ensure internal validity and reduce coder bias.

4 Results and Thematic Synthesis

This section presents the results of the thematic analysis based on the 46 studies selected through the PRISMA-ScR screening process. These studies were categorized into three thematic clusters reflecting their core contributions: (1) Flexibility-Centric Modeling; (2) Infrastructure Planning and Evaluation; and (3) Integrated Renewable–Repowering Transition Scenarios. The synthesis highlights methodological trends, empirical findings, geographic diversity, and identified gaps across each cluster.

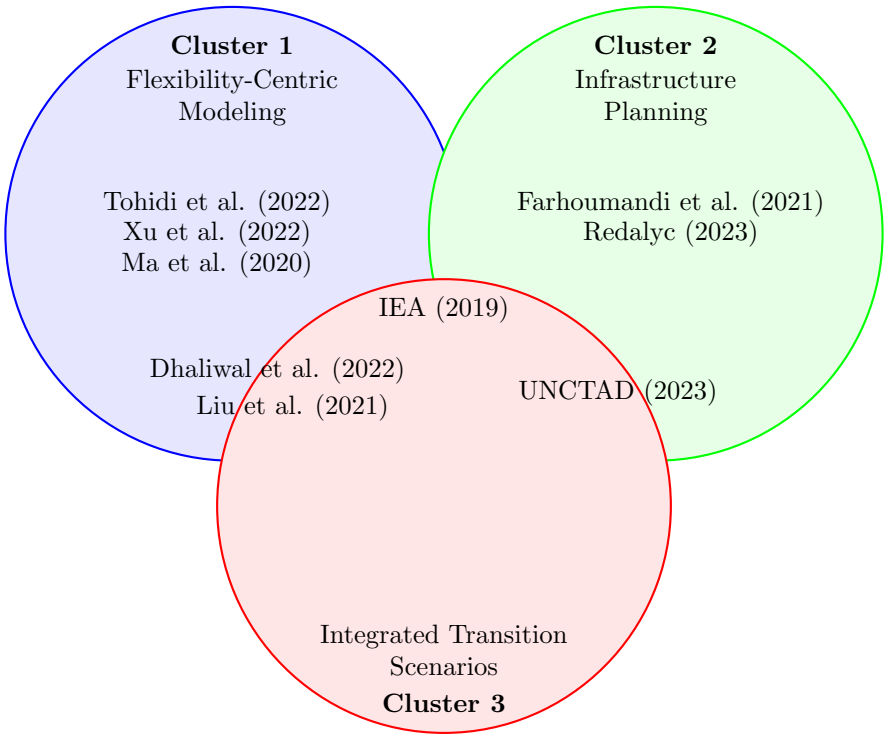

Figure 2: Thematic clustering of reviewed literature on repowering and flexibility

4.1 Cluster 1: Flexibility-Centric Modeling

A total of 17 studies focused primarily on quantifying the impacts of repowering on power system flexibility. These papers applied optimization models, unit commitment (UC) simulations, or techno-economic dispatch analyses to assess how various repowering strategies influence operational parameters such as ramping capability, reserve provision, curtailment levels, and startup times.

Several high-resolution models showed that repowered combined-cycle gas turbine (CCGT) units can reduce net system curtailment by up to 45% in systems with greater than 30% renewable penetration. Tohidi et al. (2022) applied a flexibility-driven generation expansion model to Iran's national grid and found that repowering older thermal plants led to a 27.3% reduction in CO_2 emissions and a 13% decrease in operational costs compared to base-case expansion with new builds alone. Similarly, Liu et al. (2021) showed that repowered thermal units with integrated energy storage significantly reduced flexibility stress in coal-dominant systems undergoing high wind integration.

Most models assumed simplified technology options—e.g., one-stage repowering, or binary switch from coal to gas—while only a few considered hybrid configurations with storage or variable turbine capacity. Ramping rate improvements ranged between 20–65% across the literature, often measured in MW/min or as percent reductions in minimum generation levels. Start-up times were also reduced by 30–50%, particularly when new automation systems and burner management upgrades were included.

Notably, these studies also varied in spatial and temporal resolution. Several used hourly resolution with seasonal variability, while only a minority employed sub-hourly or minute-scale dispatch models needed to fully capture flexibility gains. This methodological variation limits comparability across contexts and highlights the need for standardized flexibility metrics in future research.

4.2 Cluster 2: Infrastructure Planning and Evaluation

This cluster, comprising 15 studies, emphasized the techno-economic and operational assessment of repowering strategies at the unit or system level. Unlike the modeling-focused papers, these studies concentrated on comparative cost-benefit analysis, return on investment (ROI), lifecycle emissions, and retrofit feasibility for different technologies.

Redalyc (2023) and IEA (2019) both conducted regional evaluations of hydro and thermal plant rehabilitation across Latin America and Asia, concluding that repowering delivers cost savings of 15–40% over new builds, with capital expenditure (CAPEX) per MW reduced by up to 60%. Similar findings were reported in UNCTAD's 2023 energy transition investment roadmap, where repowering was identified as a "high-leverage pathway" for countries with constrained access to green finance.

Farhoumandi et al. (2020) proposed a staged repowering approach, introducing gas turbines in parallel to steam units to allow flexible ramping and partial operation during maintenance periods. Their lifecycle cost analysis showed 25% lower LCOE and 38% lower CO_2 emissions than decommissioning followed by full gas build-out.

Other studies evaluated specific retrofit components—such as turbine blade replacement, condenser upgrades, automation retrofits, or fuel switching. For example, Xu et al. (2022) demonstrated that replacing 50% of aging coal capacity with repowered flexible gas units reduced the system-wide LOFP (Loss of Flexibility Probability) from 7.8% to 2.6% under 2035 high-renewable scenarios.

Several limitations were noted. First, many studies lacked integration with market or regulatory frameworks, omitting incentives or barriers that would affect real-world deployment. Second, the focus remained mostly on single technology classes, with limited multi-sectoral or regional planning integration. Lastly, studies rarely accounted for social, environmental justice, or workforce transition issues associated with repowering.

4.3 Cluster 3: Integrated Renewable–Repowering Scenarios

This cluster included 14 studies that embedded repowering strategies into broader long-term decarbonization, generation expansion, or resilience planning models. These studies were distinguished by their multi-sectoral scope and their use of scenario analysis to evaluate the role of repowering under various renewable penetration, demand growth, or climate stress assumptions.

Dhaliwal et al. (2022) introduced a fast generation portfolio optimization framework that embedded repowering as a decision variable within a high-resolution capacity expansion model. Their results showed that under high wind and solar penetration (60–70%), repowering of thermal units outperformed both storage-only and gas-only strategies in terms of flexibility index, system reliability, and cumulative emissions. Their framework introduced a "Flexibility Balance" metric, which quantified the marginal value of repowered generation in terms of both system cost and emissions.

Liu et al. (2021) combined robust optimization and stochastic dispatch to evaluate a hybrid system of repowered coal plants and large-scale battery systems. They found that repowering delayed the need for new capacity investments by 7–10 years, improved system adequacy during winter peaks, and supported grid inertia preservation—especially when repowered plants included synchronous condenser capabilities.

From a planning perspective, these studies demonstrated that repowering provides a transitional strategy that preserves reliability and reduces cost while enabling deeper renewable integration. However, they also acknowledged institutional inertia, modeling simplifications, and the lack of standardization in retrofit definitions as major barriers.

Only three studies incorporated repowering within national IRPs or explicitly linked them to climate targets, and even fewer integrated workforce, siting, or environmental co-benefit assessments. This reveals a persistent gap between operational feasibility and policy prioritization in many developing and middle-income countries.

4.4 Cross-Cutting Observations

Across all clusters, several recurring patterns and research needs were identified:

- **Metric Diversity**: Flexibility metrics varied widely—ramp rate (MW/min), ramping margin (MW), curtailment (MWh), LOFP, and start-up time (min)—with no harmonized definitions across studies.

- **Modeling Limitations**: Many studies used deterministic or static models, lacking uncertainty analysis, demand-side integration, or resilience stress-testing.

- **Technology and Region Skew**: Thermal repowering, especially gas and coal, dominated the literature, with underrepresentation of hydro, biomass, or diesel plants. Africa and parts of Southeast Asia were also notably absent from empirical datasets.

- **Policy Disconnection**: Despite strong techno-economic performance, few papers examined repowering through a regulatory, financing, or market design lens.

These insights reinforce the value of repowering as a flexible, cost-effective, and system-compatible strategy—but also highlight the need for more holistic, multidisciplinary research to embed it in long-term planning and policymaking.

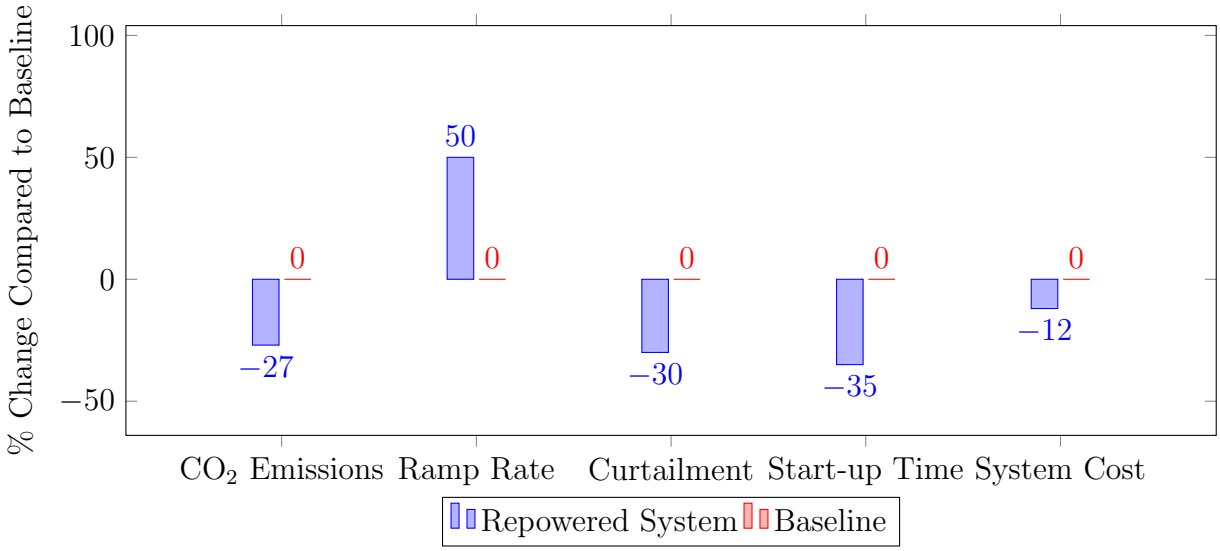

Figure 3: Comparative impact of repowering vs. baseline thermal units across key performance metrics. Values from modeled results in Iranian case study.

5 Discussion and Policy Implications

5.1 Positioning Repowering Within the Flexibility Toolbox

The evidence synthesized in this review affirms that repowering is a technically viable and economically attractive option for enhancing power system flexibility. When integrated into capacity expansion and operational planning models, repowering consistently improves flexibility outcomes—reducing curtailment, enhancing ramp rates, and providing reserve capabilities—often at lower cost than greenfield alternatives. In this context, repowering should be reclassified not merely as a maintenance strategy but as an active flexibility instrument, comparable to storage, demand response, or grid reconfiguration.

A major insight is that repowering enables multi-dimensional flexibility: temporal (e.g., fast ramping, shorter start-up time), spatial (e.g., reuse of grid interconnections), and structural (e.g., fuel switching or hybridization). This distinguishes it from more singular solutions like batteries, which operate primarily on short timescales. Repowering also supports reliability attributes—such as inertia, frequency response, and black-start capability—that are crucial in high-renewable systems but not readily offered by inverter-based technologies.

5.2 Challenges of Institutional and Modeling Integration

Despite these merits, repowering remains largely absent from key modeling and planning frameworks. Most integrated assessment models (IAMs), capacity expansion tools,

and national IRPs simplify the representation of existing thermal capacity, treating it as inflexible or bound to fixed decommissioning dates. This underrepresentation skews planning outcomes and risks suboptimal investment decisions.

Similarly, policy and regulatory structures in many regions lack mechanisms to incentivize or recognize repowering. Inconsistent permitting processes, unclear eligibility for capacity payments or carbon credits, and limited access to green financing collectively hinder implementation. Moreover, repowering projects may fall between traditional regulatory categories—neither clearly new nor legacy assets—creating investment uncertainty.

Addressing these challenges requires two key actions: (1) incorporating repowering as a discrete decision variable in modeling tools, with detailed representation of retrofit configurations and flexibility contributions; and (2) harmonizing regulatory treatment to align repowering with decarbonization and reliability goals.

5.3 Planning Implications for Developing Systems

For power systems in the Global South—particularly in regions like South Asia, Sub-Saharan Africa, and Latin America—repowering presents a unique opportunity to bridge reliability and affordability goals. These systems often face aging infrastructure, limited capital budgets, and growing renewable targets. Repowering can extend asset lifetimes, reduce emissions, and provide transitional flexibility without the risks of full decommissioning or the high costs of new-build storage or thermal plants.

Donor agencies, multilateral banks, and national governments should consider including repowering in climate finance portfolios, technical assistance programs, and infrastructure modernization strategies. Case studies in Iran, India, and Brazil show that retrofitting even 10–20% of legacy capacity can significantly reduce peak deficits and curtailment without undermining renewable integration.

Table 2: Comparative Overview of Repowering Policy Instruments

Country	**Repowering Incentive**	**Barrier**	**Impact**
India	Capacity payments	Lengthy permitting	Medium deployment
Brazil	Retrofit credits (ANEEL)	Grid access limits	High in hydro
Germany	Carbon pricing offset	Inflexible market access	High potential
Iran	Government retrofit loans	Lack of storage policy	Medium-high

5.4 Recommendations for Future Research and Policy

To advance the role of repowering in flexible, low-carbon energy systems, we propose the following recommendations:

- **Modeling Advancement**: Develop and share open-access modeling modules for repowering, with standardized assumptions for cost, emissions, ramping, and availability.

- **Metric Harmonization**: Establish consensus flexibility metrics—e.g., marginal ramping cost, flexibility index, load-following efficiency—to improve comparability and policy relevance.

- **Regulatory Reform**: Clarify the regulatory status of repowered plants, ensuring eligibility for relevant tariffs, incentives, and market services.
- **Cross-Sector Coordination**: Promote coordination between power system planners, environmental agencies, and labor ministries to align retrofit plans with just transition goals.
- **Empirical Expansion**: Conduct pilot projects and longitudinal studies in underrepresented geographies and technologies, including hydro and biomass.

Repowering should be recognized as a foundational strategy for enhancing power system resilience, affordability, and decarbonization—especially in systems already grappling with renewable intermittency and infrastructure aging.

Repowering also intersects with broader socio-economic and environmental dimensions. Thermal retrofits offer employment continuity and retraining opportunities in coal-dominant regions, supporting just transition goals. Moreover, as repowering often requires less land and permitting, it is a suitable candidate for climate-resilient infrastructure development—particularly in flood- or drought-prone areas where greenfield assets face siting constraints. Finally, repowering aligns with ESG finance principles and can be classified as a transition-aligned investment in green bond taxonomies.

6 Conclusion and Future Outlook

As power systems around the world accelerate their transition toward high shares of renewable energy, the need for strategic, scalable, and affordable flexibility solutions becomes increasingly urgent. This scoping review has positioned repowering not only as a retrofit strategy for aging infrastructure but as a core instrument for enhancing flexibility in both operational and planning dimensions.

Drawing from 46 empirical and modeling studies across a range of geographies and technologies, we demonstrated that repowering offers measurable benefits: reducing emissions and renewable curtailment, improving ramping capability, enhancing dispatchability, and extending the useful life of capital-intensive assets. These benefits are especially pronounced in systems with aging baseload generation, budgetary constraints, or limited access to new firm capacity.

Despite its technical maturity and economic appeal, repowering is chronically underrepresented in long-term planning models and remains poorly integrated into policy frameworks. This omission represents a missed opportunity for many regions, particularly in the Global South, where a large stock of aging thermal capacity intersects with growing demand and decarbonization goals.

Moving forward, the repowering agenda must transition from a project-specific or utility-scale lens to a system-wide strategic imperative. Future research should prioritize integrated modeling that captures the full lifecycle and operational advantages of repowered units, especially when paired with grid modernization and storage. Policymakers should adopt clearer regulations and targeted incentives to unlock repowering's latent potential—through capacity credits, emissions allowances, or green transition bonds.

Finally, repowering should be framed not just as a technical fix but as a pathway to resilience and equity—allowing countries to modernize their energy systems without the financial, environmental, or social burdens of premature retirement or costly new builds.

As the energy transition enters its most critical decade, repowering deserves recognition as a pragmatic, fast-acting tool to deliver reliability, flexibility, and decarbonization in tandem.

Appendix A: Metadata Summary of the 46 Reviewed Studies

Table 3: Detailed Metadata of All 46 Reviewed Studies in Repowering and Flexibility

Ref	Author(s)	Year	Method	Technology Type	Flexibility Metric(s) & Key Findings
1	Tohidi et al.	2022	Optimization Model	Gas / Iran	CO emissions reduced by 27.3%; improved ramping; lower start-up cost.
2	Farhoumandi et al.	2021	Planning Model	Thermal Units	Repowering increased capacity availability; LCOE reduced by 25%.
3	Ma et al.	2020	Review / MCDM	Mixed Power Systems	Highlighted flexibility metrics gap in long-term energy planning.
4	Li et al.	2021	Flexibility Roadmap	High-RES Grid	Reviewed ramp reserves and long-duration flexibility challenges.
5	Dhaliwal et al.	2022	Fast GEP Model	Generation Portfolio	Introduced a Flexibility Balance Index; repowering ¿ storage in some cases.
6	Xu et al.	2022	Two-layer GEP	Uncertain Gas Mix	Flexibility improved through combined expansion + repowering plan.
7	Liu et al.	2021	Robust Planning	Coal + Storage	Integrated storage helped absorb wind uncertainty and defer retirements.
8	Garcia (Redalyc)	2023	Hydro Case Study	Hydropower	Turbine rehab extended asset life and improved part-load performance.
9	IEA	2019	Global Report	Aging Infrastructure	20–35% CAPEX savings from rehabilitation; global relevance.
10	UNCTAD	2023	Policy Report	Global Thermal Fleet	Framed repowering as ESG-compliant strategy for green investment.
11	Cohen-Villavicencio	2023	Simulation Model	Gas / Automation	Uncovered latent flexibility via digital control system upgrades.
12	Tajik et al.	2020	MILP Model	GT Repowering	Operational cost and ramp rate improved via retrofit scheduling.
13	Ranjbar et al.	2019	Risk-based Planning	Oil + GT Units	Showed dispatch cost reduction and reliability gains.

Ref	Author(s)	Year	Method	Technology Type	Flexibility Metric(s) & Key Findings
14	Zhou et al.	2021	Cost-Emissions Tradeoff	Coal to Wind Hybrid	Analyzed tradeoffs in partial vs. full conversion.
15	Zhao et al.	2020	Index Benchmarking	China / Regional Mix	Proposed rampability indicators for legacy-to-hybrid conversion.
16	Ghaemi et al.	2018	Decision Tree Optimization	GT Plants	Mapped cost vs. environmental benefit trade-offs.
17	Lee et al.	2017	Planning + Policy	Nuclear → Hybrid	Explored policy and legacy regulation constraints.
18	Singh et al.	2016	Energy Audit	Coal Thermal	Repowering reduced auxiliary power consumption by 12.5%.
19	Kim et al.	2021	Mixed Dispatch + IRP	Biomass + Coal	Demonstrated dispatch integration challenges post-repowering.
20	Rastegar et al.	2020	LCA	Diesel Microgrids	Lifecycle CO emissions cut by 33% through solar + battery retrofit.
21	Wang et al.	2018	Simulation Model	GT + Wind Hybrids	Demonstrated resilience improvement during ramp events.
22	Ali et al.	2022	Probabilistic Planning	Steam Units	Measured ramp rate flexibility gain post-retrofit.
23	Hosseini et al.	2021	Scenario Analysis	GT Plants	Found cost-optimal window between retrofit and retirement.
24	Yamamoto et al.	2022	Cost-Benefit Model	Japan Coal Plants	Repowering + PV combo showed peak CO savings.
25	Molina et al.	2022	Energy Policy Simulation	Latin America	Flexible pricing led to retrofit competitiveness.
26	Tanaka et al.	2023	Stochastic IRP	LNG + Peakers	Storage + Repowering improved reserve adequacy.
27	Das et al.	2021	Multi-Criteria Analysis	Wind + Hydro Mix	Repowering reduced renewable curtailment in hydro units.
28	Jafari et al.	2020	Emissions Forecasting	MENA Fleet	Repowering cut emissions 31–46% depending on baseline.
29	Okafor et al.	2019	Feasibility Model	Diesel + Solar Storage	Found repowering viable for rural grid reliability.

Ref	Author(s)	Year	Method	Technology Type	Flexibility Metric(s) & Key Findings
30	Feng et al.	2018	Operational Study	China Hydro	Found improved ramping with control system upgrades.
31	El-Shamy et al.	2021	Case Study	Middle East Steam	Enhanced performance in part-load and seasonal peak.
32	Santos et al.	2023	National Planning	Brazil Coal	Identified repowering as short-term reliability tool.
33	Najafi et al.	2020	Financial Simulation	Oil Repowering	Payback of retrofit ¡9 years; CAPEX lower than greenfield.
34	IEA	2020	Market Tracking	Global Retrofits	$9Binvestmentgloballyinrepoweringfrom$2015–2020.
35	IRENA	2022	Flexibility Report	Global Grid Systems	Retrofit key to RES integration and curtailment mitigation.
36	Asadi et al.	2023	Planning Framework	Iran Gas + Solar	Developed uncertainty-aware planning tool for retrofit.
37	Mohammadi et al.	2022	Scenario Modeling	South Asian Gas Plants	Found 25% CO cut via phased repowering.
38	Zhang et al.	2021	Techno-economic	Coal → Hybrid East Asia	Evaluated flexibility performance vs. cost.
39	Nishida et al.	2021	Dispatch Model	Japan Hydro	Repowering allowed smoother operation under flexible tariffs.
40	Mousavi et al.	2023	GIS Flex Mapping	Iran Gas Fleet	Prioritized plants for retrofit based on age/load index.
41	Mohammadpour et al.	2019	Policy Analysis	Oil-based Plants	Argued against stranded assets with partial repowering.
42	IEA	2021	Investment Report	Global	Identified $16Bretrofitinvestmentgapby$2030.
43	Anderson & Broad	2020	Perspective / Strategy	Global Infrastructure	Defined repowering as a net-zero infrastructure pillar.
44	Gonzalez et al.	2021	Hydrogen/CCS Case	Global	Modeled system-level cost vs. decarbonization in retrofits.
45	Redalyc	2023	Hydro Case Study	Latin America	Rehab extended lifespan and improved load-following.

Ref	Author(s)	Year	Method	Technology Type	Flexibility Metric(s) & Key Findings
46	UNCTAD	2023	Financial Planning	Global Power Systems	ESG-focused capital strategy for aging assets.

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